

Modeling, Animation, and Rendering of a 3-Dimensional Object

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Introduction: For this project, I had the idea of working with a reusable model so I could make my scene more interesting without having to render too many different objects. My goal was to implement the Blender techniques we learned in our activities in order to add more realistic attributes to the scene, as well as my own touch of some fun sound effects. Since I am from Boston, my mind (for some odd reason) jumped to *Make Way for Ducklings* by Robert McCloskey and I envisioned a line of ducklings led by their mother on a sunny day, so I chose to model a rubber duck. I scaled the duck so that the mother was larger than her three ducklings and animated them so that they followed her along in the water. Since this is not too interesting, I decided on adding a third object, a boat passing by on the water. This was a fun part of the project in which I collaborated with one of my friends and was able to 3D-print a boat model and take pictures and videos of it for modeling purposes.

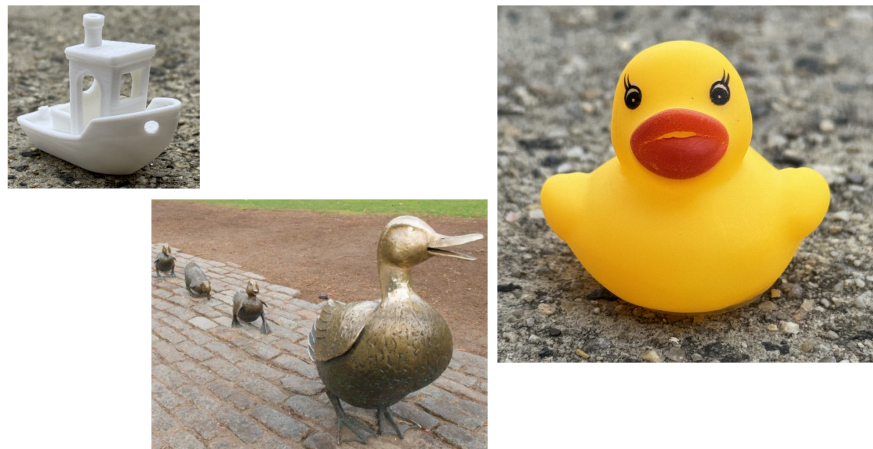


Figure 1

Literature Review: Though I did not use as much literature this year as I have on past projects, a few sources which were helpful to me were *Research on the production process of digital media 3D animation: Taking Blender as an example.* by Chen, C. (2024), *Progressive 3D reconstruction of infrastructure with videogrammetry* by Brilakis et al. (2011), as well as the Blender Fundamentals YouTube channel tutorials, which helped me greatly in understanding the ins and outs of a

realistic water look as well as giving clear answers to many of my animation questions, especially when optimizing rendering speeds. The first article by Chen (2024) laid out a succinct overview of the step-by-step processes which go into animating a 3D model. This article provides an excellent overview of the goals of using Blender to represent real world objects. Since I attempted both photogrammetry and videogrammetry using Meshroom and the Scaniverse mobile software, I became interested in what went on behind the scenes in the formation of point clouds from a simple video file parameter. The article by Brilakis et al. (2011), though it is slightly older, provided some relevant insights into the applied concepts of this process and how a software would operate with these specific capabilities. Though I did not dive too deeply into all the specifics, I was interested to see the different applications of this kind of software in other fields of engineering, such as civil and biomedical engineering.

Methodology: I began this unusually eventful but surprisingly fun project by selecting both the rubber duck and 3D-printed boat objects in order to transform them into virtual 3D objects for my animation. My first approach was to download and utilize the Meshroom and Meshlab softwares to model and touch up on my object's stray vertices. This ended up proving very difficult, but I continued to test different lighting environments and locations including an overcast day with essentially negligible shadowing. I took around two thousand photos in all (several attempts at rendering each object) and most of my results proved to be choppy with stray or missing vertices. Figures 1-4 represent some of my trial scans of my duck and my 3D-printed boat using photogrammetry in Meshroom and editing in Meshlab:

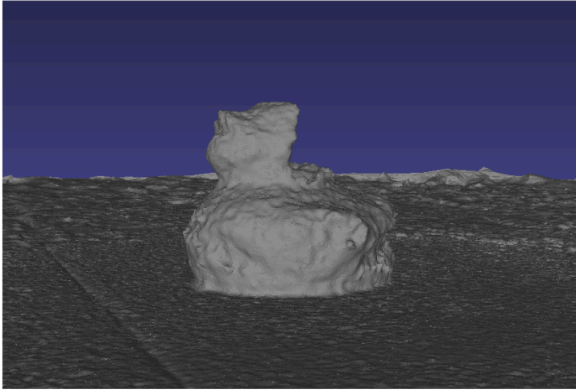


Figure 2

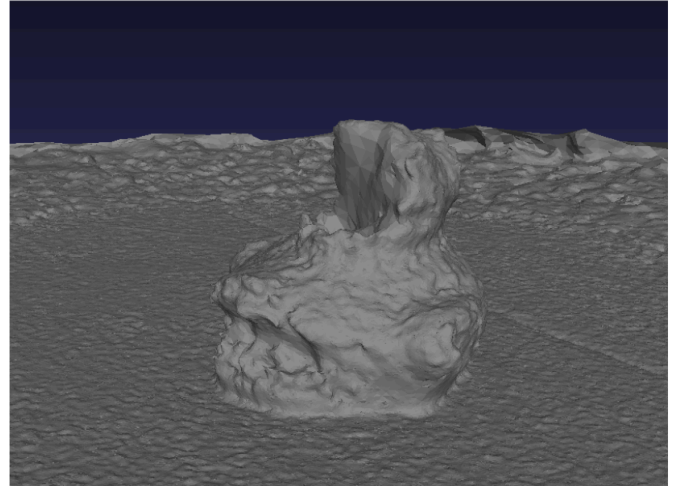


Figure 3

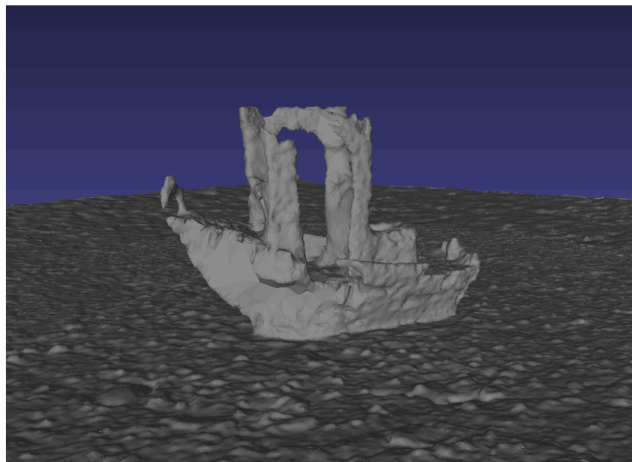


Figure 4

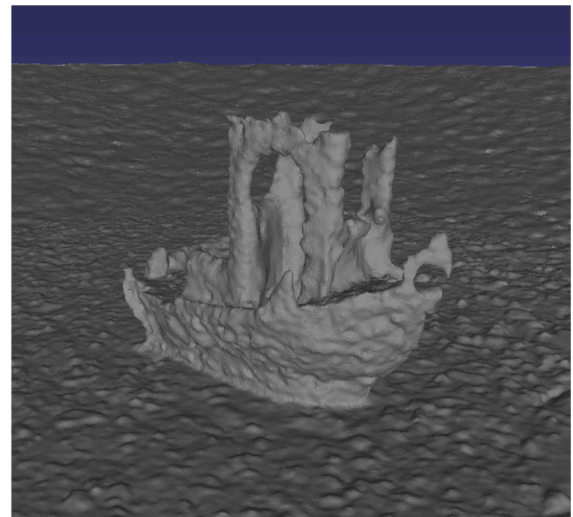


Figure 5

Though I tried many things, such as importing into Blender and sculpting, I was unable to get the smooth model that I was looking for. I decided to use a different mobile software called Scaniverse to see if I could capture some better results. I used the built-in videogrammetry software on a statue I had in my room as an initial test of the software's accuracy and this was my result:



Figure 6

I was very happy with how my model in Figure 6 had turned out and resolved to try using it on my duck and sculpting the rough edges in Blender to achieve my desired look. After a few attempts at using the Scaniverse modeling software, I was able to get a fairly accurate result and decided to import my file into Meshlab first and then a Blender workspace to edit it up. Using my knowledge of Blender, I entered the sculpting and texturing windows and was able to create a fairly realistic 3D model of my rubber duck which I was happy with. Here is the initial videogrammetry scan result of my textured, but unsculpted duck model in Meshlab, where I removed the excess base vertices, before importing it to Blender:



Figure 7

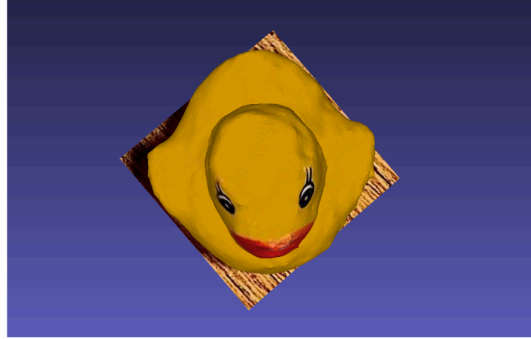


Figure 8

I was not quite happy with the bumps on the duck's surface, so I went into the Blender sculpting window and touched it up. Figure 9 displays the duck model in a lighting rendered environment and Figure 10 displays a side view of the final model without render lighting in the 3D viewport:



Figure 9

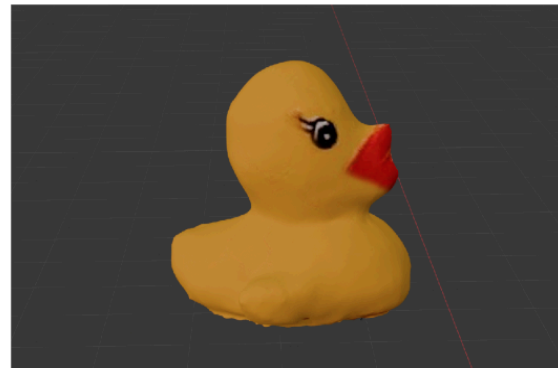


Figure 10

Now, once I had completed this long and tedious sculpting process, I took to working on my boat model. After many, many scans I was unable to get a good render of the detail I wanted on my 3D-printed boat. This was a big disappointment, since I had collaborated with my friend to print out the model, but ultimately I decided to simply import the .stl model I had used to print the boat into Blender and to stick with only one 3D model, the duck. Figures 11 and 12 demonstrate just a few of my many failed attempts at modeling my 3D-printed boat:



Figure 11



Figure 12

When I was satisfied with my models, I took to creating my scene in Blender. This part was fun, but challenging. One challenge I faced was very frustrating, I deleted my entire file after working for hours on it. This was my own fault due to file management issues, but I got back on track, trying to remember the techniques I had implemented. My first technique, from the Blender activities, was to import a suitable HDRI image which would act as a backdrop for my lake. I loved the idea of having a waterfall and found a great image which would couple nicely with the waterfall sound I found later on in the process. I added some bushes which I imported to give the scene a more natural look and generated an appropriate light source, just about central to where the pond would be. Then, I added a plane and properly textured it with a greenish-blue, reflective material. Next, I implemented some standard water motion in the physics tab, subdividing my plane into a canvas and making all other objects which touched it into brushes that acted on the water. Figure 13 shows a snapshot of the physics tab for my plane, which acted as the pond:

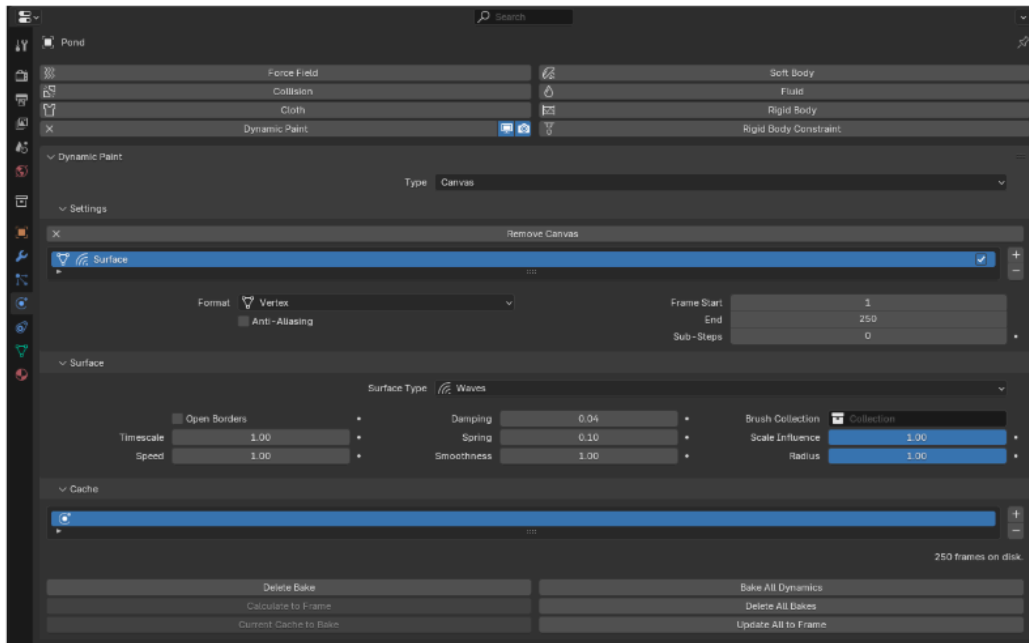


Figure 13

Now that I had my scene set in place and had tested the water physics, I was ready to begin my animation. This was my favorite part of the project, though it was admittedly difficult to get all the ducklings to move in the way that I wanted. I attempted to use Bezier curves at first to make my ducks follow a loop pattern, but eventually settled on a straight path motion with rotation, for a more natural look. I synchronized the boat's movement with the turning of the ducks to give a spooked effect when the horn goes off in the animation. I made each of the ducklings move in a slightly different path, to give a smooth effect when they all turned to look at the camera before swimming off, the littlest duckling trailing slightly behind the rest. Once I was satisfied with my animation, I began the rendering process. At first, I was shocked that it was taking so long to render (almost 5 minutes per frame), but after switching a few settings, directing the processing power to my GPU, making changes to my subdivision surface modifiers, and scaling down the way that the light was rendered, I was able to significantly reduce the time it took to render my animation into .png image frames. I then compiled these .png files in the very convenient

Blender video editing software window, where I synchronized fun, open source sound effects with the movement of my objects to make my scene come alive. Once I had downloaded the final .mp4 file, I uploaded the roughly 15 second video to YouTube with a custom title, thumbnail, and project description.

Results: The final animation came out well, I was happy with the way the ducks looked on the water and how they moved together. I was also very happy with the boat's movement and the way that it caused a wake on the water. It is not clearly visible in the video, but the boat rotates slightly while creating a wake, giving a realistic effect. I have supplemented my report with Figure 14 to show this aspect of my animation and to provide a better viewing angle:



Figure 14

Discussion: I was able to apply photogrammetry and videogrammetry concepts to a real-world object of my choosing and was also able to interconnect different concepts we had learned in our course to the field of animation. To understand how the GPU and CPU operate in computing millions of matrix multiplications and transformations every second is key to getting a glimpse behind the scenes of how animation actually takes place. Though it is unlikely that I will enter

the field of computer graphics, I have undoubtedly taken away vital concepts such as the use of triangles to comprise any 3D model efficiently and effectively and the rigid transformation (rotation, scaling, shearing, etc.) and non-rigid translation using homogeneous coordinates of matrices (among others), which are fundamental to the field of computing. One thing that I was not happy with was the underside of my duck's bill, which I was not able to sculpt perfectly and which still had a visible blemish. My sculpting and texturing technique is something I can improve on in the future, though I think I did a good job of improving from my initial scan. Another aspect of the animation that would benefit from future improvement is further subdivision of the water plane to give a smoother, more realistic effect. Though not apparent at a first glance, if you look very closely at when the light hits the plane, you may notice subdivisions in the water surface which look unrealistic. For rendering efficiency purposes, I was afraid my device would have performance issues if I subdivided the surface any further, so I attempted to optimize my settings and to work with what I could realistically render. Overall, I was content with the final video and was satisfied with my first official Blender animation project (and first YouTube upload!).

Conclusion: In conclusion, I am happy to say that I gained a good deal of knowledge and experience from this project. Though the animation itself was simple and the techniques fairly basic in the grand scheme of things, I was able to experiment with many different ways of approaching the modeling and animation process. It is important to have these foundational skills and to be able to abstract principles of computing which can be used later on. I enjoyed the freedom of being able to choose what to model as this gave the project a lightness and ease which made creativity more accessible.

Appendix

Brilakis, I., Fathi, H., & Rashidi, A. (2011). Progressive 3D reconstruction of infrastructure with videogrammetry. *Automation in Construction*, 20(7), 884–895.

<https://doi.org/10.1016/j.autcon.2011.03.005>

Chen, C. (2024). Research on the production process of digital media 3D animation: Taking Blender as an example. *Highlights in Art and Design*, 7(2), 28–31.

<https://doi.org/10.54097/xm4cba08>

Blender. (n.d.). *Home [YouTube channel]*. YouTube. Retrieved April 30, 2025, from

<https://www.youtube.com/@BlenderOfficial>



YouTube Link:

<https://youtu.be/arMj4OzClVo>



GitHub Link:

<https://github.com/ndalsass/CSC322Spring/tree/main/FinalProject>